ModelIt!

Student Activity Pack

Powered by the Sun

How Pizza, Basketball, and Your Heartbeat Are All Solar Energy

Grade 5 | NGSS 5-PS3-1 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where does the energy in your food come from? Trace it back as far as you can.

2 Why do you think bears hibernate in winter but not summer?

3 Draw an arrow showing how the sun's energy gets into fruit you eat.

4 What would happen to animals if all plants suddenly disappeared?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sunlight Intensity | Temperature | Photosynthesis Rate | Energy Available to Animals

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Summer Conditions

High sunlight, warm temperature

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Winter Simulation

Lock both Sunlight and Temperature to 25%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Eclipse/No Sun

Lock Sunlight to 0%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Water — Plants need water for photosynthesis

CO2 — Carbon dioxide is the raw material plants use

Herbivore Population — Number of animals eating plants

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does energy from the sun become the energy in your body? Use evidence from your simulation to support your answers.

1 Why do ecosystems in tropical regions support more animal life?

2 Why do many animals store fat before winter?

3 If the sun provided half as much energy, what would Earth look like?

4 How is a fossil fuel like coal actually stored solar energy?

5 Trace the energy in something you ate today back to the sun.

8 STEM Challenge: Design a Greenhouse for Mars

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YOUR ENGINEERING MISSION

Design a greenhouse that could grow food for astronauts on Mars, where sunlight is weaker.

THE SCENARIO

NASA is planning a Mars colony. They need you to design a greenhouse that maximizes photosynthesis with limited sunlight.

Engineering Design Thinking

1. WHAT LIGHT SOURCES WILL YOU USE?
2. HOW WILL YOU CONTROL TEMPERATURE IN THE HARSH MARTIAN ENVIRONMENT?
3. WHICH CROPS GROW BEST IN LOW-LIGHT CONDITIONS?
4. HOW WILL YOU CONSERVE ENERGY WHILE MAXIMIZING PLANT GROWTH?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Sports Nutritionists and Exercise Physiologists optimize how athletes convert food energy into performance. They earn $55,000-$90,000/year.